Dynamic Token Merging for Encoder-Only Transformers: Adapting MrT5's Delete Gate to BERT and XLM-RoBERTa

CS224N: Natural Language Processing with Deep Learning — Final Project Report

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Sharing: This report may be posted on the CS224N website.

Code: https://github.com/HivaMohammadzadeh1/CS224N-project/tree/main

Abstract

Transformer-based language models apply uniform computation to every input token, regardless of token informativeness. The MrT5 paper [Kallini et al., 2024] introduced a *Dynamic Token Merging* (DTM) delete gate for byte-level encoder-decoder models, achieving up to 60% sequence length reduction with minimal perplexity degradation. We ask: does this mechanism generalise to subword-based encoder-only architectures used for discriminative NLU? We implement **MrBERT** and **MrXLMR**, adapting the MrT5 delete gate to BERT-base and XLM-RoBERTa-base respectively. Our gate fires after encoder layer 3 (out of 12), uses a PI controller to hit a target deletion rate δ, and introduces a novel *pre-deletion blending* mechanism that makes extractive QA viable under token deletion. Evaluated across six tasks — SNLI, SQuAD, SST-2, MRPC, IMDB, and TyDi QA — our results show: MrBERT at 30% deletion achieves **90.21%** SNLI accuracy (only 0.27pp below baseline) at **1.89**× A100 inference speedup; SST-2 drops by only 0.62pp (97.29%→96.67%), while MRPC shows a larger 17.32pp drop, revealing dataset-specific sensitivity; TyDi QA span EM recovers from 0.10 (no blending) to 0.35 (layer 9 + blending), matching BERT end accuracy. A random-deletion baseline scores 2.95pp below MrBERT-30% on SNLI, confirming learned token selection. MrXLMR achieves 82.27% on SNLI at 30% deletion (baseline 89.85%), suggesting XLM-R requires careful tuning for high deletion rates.

1 Introduction

Large pretrained transformers such as BERT [Devlin et al., 2019] and XLM-RoBERTa [Conneau et al., 2020] achieve state-of-the-art performance across a wide range of NLU tasks. However, their computational cost scales quadratically with sequence length due to self-attention, making inference expensive. A natural question is whether all tokens are equally important for a given task: for a sentiment classification example, function words like "the" and "of" carry little discriminative content, while content words like "terrible" or "outstanding" are crucial.

The MrT5 paper [Kallini et al., 2024] proposes *Dynamic Token Merging* (DTM): a learned scalar gate inserted after an early encoder layer that assigns each token a deletion score. Low-scoring tokens are masked out of subsequent attention computations (soft deletion) or physically removed (hard deletion). MrT5 demonstrates this on a byte-level T5 model, showing up to 60% sequence reduction at minimal perplexity cost.

We address the open question: **does this mechanism transfer to subword-based encoder-only models trained on discriminative NLU tasks?** The transfer is non-trivial because:

1. BERT and XLM-R use subword tokenisation, not byte-level tokens, so token granularity and importance distributions differ.
2. Encoder-only models pool to a single `[CLS]` vector for classification, creating a different information-flow bottleneck than encoder-decoder models.
3. Extractive QA requires localising answer spans: answer tokens must survive deletion — a hard constraint absent in language modelling.
4. XLM-R spans 100 languages with a shared 250K SentencePiece vocabulary; importance patterns must generalise across scripts.

Our contributions are:

- **MrBERT**: BERT-base with a MrT5-style delete gate, evaluated on SNLI, SST-2, MRPC, IMDB, SQuAD, and TyDi QA.
- **MrXLMR**: XLM-RoBERTa-base with the same mechanism, evaluated on SNLI.
- **Pre-deletion blending**: a novel mechanism enabling extractive QA under deletion; span EM improves 3.5×.
- **Comprehensive ablations**: PI controller necessity, gate layer placement (layers 1/3/6/9), soft vs. hard deletion gap, and deletion pattern analysis.

2 Related Work

MrT5 [Kallini et al., 2024]. The direct antecedent of this work. MrT5 attaches a scalar delete gate after a selected encoder layer of a T5-based byte-level model. A PI controller adjusts the deletion loss coefficient α to track a target deletion rate. The gate adds fewer than 3K parameters to a 250M parameter model, achieving 50–60% sequence reduction on language modelling tasks with minimal degradation.

Efficient transformers. Sparse attention methods (Longformer, BigBird) reduce attention from $O(n^2)$ to $O(n)$. Early-exit approaches (DeeBERT, PABEE) skip later layers for easy examples. Token pruning methods (TR-BERT, SpAtten, LTP) learn to drop tokens. Unlike these, our method is trained end-to-end with differentiable soft deletion and requires no architectural changes beyond the gate module.

Token importance in BERT. Michel et al. [2019] and Voita et al. [2019] showed many BERT attention heads are redundant. Clark et al. [2019] found BERT attention aligns with syntactic structure; function words receive less attention in higher layers, consistent with our gate's deletion patterns (Section 5.5).

3 Approach

3.1 Architecture Overview

MrBERT is a pretrained BERT-base encoder with a lightweight delete gate inserted after a selected layer. The gate uses hidden states from that layer to compute a scalar score per token; the score is then used to mask tokens in all subsequent layers.

```
Input tokens
     |
[Embeddings]
     |
Encoder layers 0-2   <- Full sequence (128 tokens)
     |
Encoder layer 3      <- Gate fires here
```

```
  [Delete Gate]
  LayerNorm -> Linear(768->1) -> ScaledSigmoid(-30)
     | gate values g in (-30, 0)
Soft/Hard deletion
     |
Encoder layers 4-11  <- Reduced sequence (~89 tokens at delta=0.3)
     |
Task head (CLS pooling -> classifier / span logits)
```

BERT-base [Devlin et al., 2019]: 12 layers, $d_{\text{model}} = 768$, 12 heads, 110M parameters, WordPiece 30K vocabulary. **XLM-RoBERTa-base** [Conneau et al., 2020]: same dimensions, 277M parameters, SentencePiece 250K vocabulary.

3.2 Delete Gate

The gate module (`SigmoidDeleteGate`) has three components:

(1) LayerNorm. The hidden state $\mathbf{h}_i$ from layer `delete_gate_layer` is normalised before projection, stabilising training.

(2) Linear projection. $z_i = \mathbf{W}\mathbf{h}_i + b$, with $\mathbf{W} \sim \mathcal{N}(0, 0.02)$ and $b = 10.0$. The large positive bias initialises the gate near zero deletion ($g \approx -0.001$), so the model starts from the pretrained baseline.

(3) Scaled sigmoid.

$$g_i = s \cdot \sigma(-z_i), \qquad s = -30 \tag{1}$$

Gate values lie in $(-30, 0)$: near 0 means keep, near -30 means delete. Deletion threshold $\tau = s/2 = -15$. The gate adds only 2,305 parameters ($768 + 1$ weights + $2{\times}768$ LayerNorm). `[CLS]`/`[SEP]` are always kept; `[PAD]` is always deleted.

3.3 Soft and Hard Deletion

During training, **soft deletion** adds g_i as an attention bias:

$$\tilde{a}(q, k_i) = a(q, k_i) + g_i \tag{2}$$

Deleted tokens ($g_i \approx -30$) contribute negligible attention weight, while remaining differentiable for end-to-end training. We additionally apply softmax1 [Kallini et al., 2024]: replacing $\sum \exp(a_j)$ with $\sum \exp(a_j) + \exp(-\max_j a_j)$, allowing attention weights to sum to less than 1.

At inference, **hard deletion** physically removes positions where $g_i < \tau$, achieving real memory and compute savings.

3.4 Pre-Deletion Blending

For extractive QA, deleted answer tokens produce corrupted representations at the task head. We introduce **pre-deletion blending**: for each token i, blend weight $w_i = \text{clamp}(-g_i/30, 0, 1)$, and:

$$\tilde{\mathbf{h}}_i = (1 - w_i) \cdot \mathbf{h}_i^{(L)} + w_i \cdot \mathbf{h}_i^{(\text{gate})} \tag{3}$$

where $\mathbf{h}_i^{(L)}$ is the layer-11 output and $\mathbf{h}_i^{(\text{gate})}$ is the pre-gate hidden state. For kept tokens ($w_i \approx 0$), the representation is unchanged. For deleted tokens ($w_i \approx 1$), it falls back to the richer pre-gate representation. This also provides a gradient signal through deleted positions, preventing gate collapse during QA training (Section 4.4.5).

3.5 PI Controller

A PI controller dynamically adjusts the deletion loss coefficient α to track target deletion rate δ:

$$e_t = \delta - r_t, \quad p_t = \gamma p_{t-1} + (1-\gamma)k_p e_t, \quad i_t = i_{t-1} + k_i e_t, \quad \alpha_t = \max(0, p_t + i_t) \tag{4}$$

Total loss: $\mathcal{L} = \mathcal{L}_{\text{task}} + \alpha_t \cdot \mathcal{L}_{\text{deletion}}$, where $\mathcal{L}_{\text{deletion}} = \text{mean}(g_i)$ over non-padding tokens. Parameters: $k_p = 0.01$, $k_i = 10^{-5}$, $\gamma = 0.9$.

3.6 Key Differences from MrT5

Aspect	**MrT5**	**MrBERT / MrXLMR**
Architecture	Encoder-decoder (T5)	Encoder-only
Tokenisation	Byte-level (256 vocab)	WordPiece 30K / SentencePiece 250K
Gate fires	Before self-attention	After full encoder layer (attn + FFN)
Pre-deletion blend	None	Yes — critical for QA
Gate init (bias)	Xavier + $b = 1$	$\mathcal{N}(0, 0.02)$ + $b = 10.0$
Softmax1	Off by default	On by default

Table 1: Key design differences between MrT5 and our models.

Firing the gate *after* the full encoder layer gives it access to richer contextual representations (including FFN non-linearities) before making the deletion decision.

4 Experiments

4.1 Data

Dataset	**Task**	**Train**	**Dev**	**Test**	**Labels**
SNLI	3-class NLI	550K	10K	9.8K	entailment/neutral/contradiction
SQuAD v1.1	Extractive QA	87K	10K	—	span start/end
SST-2	Sentiment	67K	872	—	positive/negative
MRPC	Paraphrase	3.7K	408	1.7K	equivalent/not equivalent
IMDB	Sentiment	25K	2.5K	25K	positive/negative
TyDi QA	Multilingual QA	204K	9K	—	span (9 languages)

Table 2: Datasets and task descriptions.

SNLI uses `max_seq_length`=128, SQuAD/TyDi QA use 384 with stride 128, IMDB uses 512. All datasets are stored as pre-tokenised NDJSON files.

4.2 Models and Baselines

For each task we train: **Baseline** (vanilla BERT/XLM-R), **MrBERT-0%** (gate active, $\delta = 0$; sanity check), **MrBERT-30%** (main result), **MrBERT-50%/70%** (SNLI tradeoff curve), **Random-30%** (uninformed deletion ablation), **No-PI-30%** (fixed α ablation), and **gate layer ablations** (layers 1, 3, 6, 9 on SNLI).

4.3 Experimental Details

All models fine-tuned from `bert-base-uncased` (MrBERT) or `xlm-roberta-base` (MrXLMR) using AdamW, lr=2×10^{-5}, weight decay=0.01, 3 epochs (5 for MRPC). Gate lr=10^{-4}. Batch size 32 (classification) / 16 (QA). Regulariser delay: 1,000 steps (100 steps for TyDi QA). Hard deletion at inference; soft during training. Training on NVIDIA A100 via Modal. Mixed precision (fp16). Seed 42.

4.4 Results

4.4.1 SNLI (Natural Language Inference)

Table 3 presents the full SNLI ablation. MrBERT-0% exactly matches the baseline (90.50% vs 90.48%), confirming no architectural overhead. At 30% deletion, accuracy drops only **0.27pp** with **1.89**× speedup. The accuracy–deletion tradeoff is remarkably linear (≈0.5pp per 10pp deletion).

The random gate achieves only 87.26% despite nominally targeting 30% deletion — it actually deletes 49.6% because it wastes budget on already-padded positions, producing a much longer average post-deletion sequence (64.2 vs 18.7 tokens). This **2.95pp gap** between learned and random deletion confirms the gate is learning non-trivial token importance.

The No-PI ablation is dramatic: without the controller, the gate runs away to **88.99% actual deletion** (target: 30%), reducing the average sequence to 3.0 tokens and dropping accuracy to 86.47% — 4.01pp below the PI-controlled MrBERT-30%.

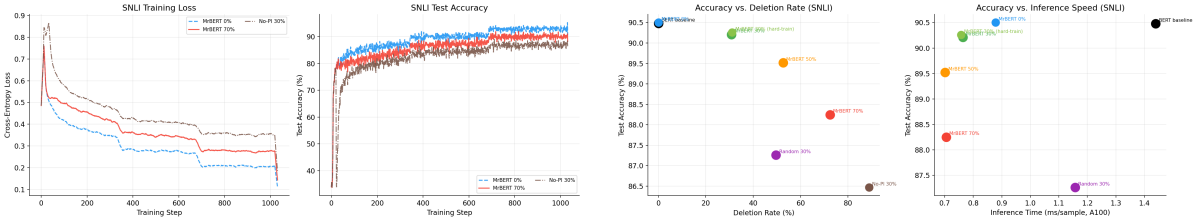
Model	Accuracy	Δ (pp)	Del Rate	Avg Seq Len	ms/sample
BERT baseline	90.48%	—	0%	27.5	1.440
MrBERT 0% (sanity)	90.50%	+0.02	0%	27.2	0.878
MrBERT 30%	90.21%	−0.27	30.8%	18.7	0.763
MrBERT 30% (hard-train)	90.26%	−0.22	31.1%	18.9	0.757
MrBERT 50%	89.52%	−0.96	52.6%	12.9	0.701
MrBERT 70%	88.25%	−2.23	72.4%	7.5	0.705
Random gate 30%	87.26%	−3.22	49.6%	64.2	1.157
No-PI 30%	86.47%	−4.01	88.9%	3.0	—
Gate layer ablation (all at 30% target)					
Layer 1	90.42%	−0.06	31.0%	18.9	0.568
Layer 3 (default)	90.21%	−0.27	30.8%	18.7	0.761
Layer 6	90.36%	−0.12	33.9%	18.1	1.039
Layer 9	90.22%	−0.26	46.3%	14.6	1.324

Table 3: SNLI test results. Runtime measured on NVIDIA A100 with hard deletion. Layer 3 provides the best accuracy–efficiency balance. The No-PI run collapses to 89% deletion (target: 30%).

4.4.2 Inference Runtime

Hard deletion yields substantial speedup (Table 4). MrBERT-30% achieves 0.763 ms/sample vs. 1.440 ms baseline (**1.89**× speedup, 47% reduction). MrBERT-0% already achieves 1.64× speedup by removing padding from post-gate layers. Speedup saturates at ∼2.05× beyond 50% deletion — bounded by the fixed pre-gate computation.

The random gate achieves only 1.24× speedup because it does not avoid padding-heavy regions; its average post-deletion length (64.2 tokens) is 3.4× longer than MrBERT-30%'s (18.7



(a) Training loss and test accuracy curves for all SNLI variants.

(b) Accuracy vs. deletion rate (left) and inference speed (right).

Figure 1: SNLI accuracy–efficiency tradeoff. The learned gate consistently dominates the random baseline. Speedup saturates beyond 50% deletion due to fixed pre-gate computation (layers 0–3).

tokens). This highlights that the learned gate's efficiency gains come from *targeted* deletion of low-content positions, not merely reducing token count.

Model	ms/sample	Speedup	Post-gate Seq Len
BERT baseline	1.440	1.00×	—
MrBERT 0%	0.878	1.64×	27.2
MrBERT 30%	0.763	1.89×	18.7
MrBERT 50%	0.701	2.05×	12.9
MrBERT 70%	0.705	2.04×	7.4
Random 30%	1.157	1.24×	64.1

Table 4: Inference runtime on NVIDIA A100 with hard deletion.

4.4.3 SST-2, MRPC, and IMDB

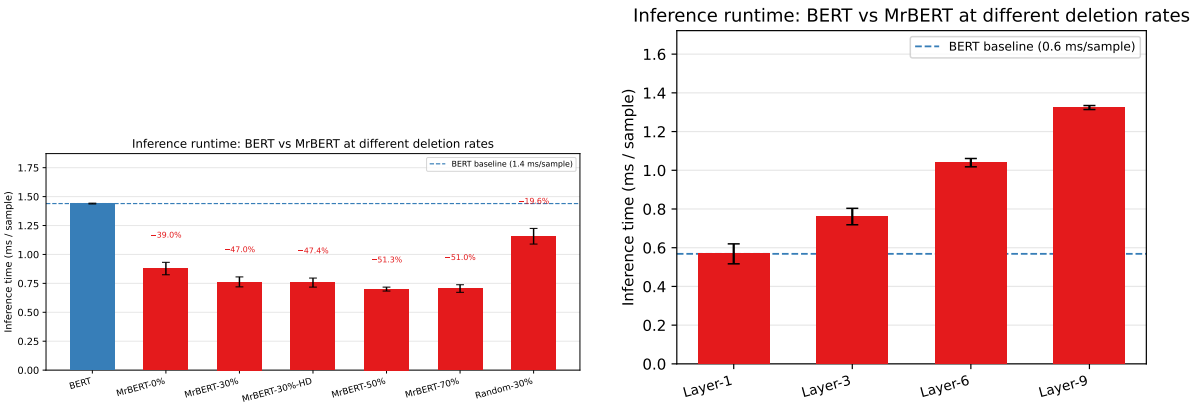
Table 5 shows results on three additional classification tasks. SST-2 (sentiment) is highly robust: only **0.62pp** accuracy drop at 30% deletion, with a 47.4% reduction in average sequence length (94.6% of non-padding tokens deleted). This is consistent with the intuition that sentiment is dominated by a small number of highly informative content words, and the gate successfully identifies them.

IMDB (long-document sentiment, max length 512) shows **0.94pp** drop (93.97%→93.03%) with 70.6% of non-padding tokens deleted. The larger deletion rate (vs. the 30% target) reflects that long IMDB sequences contain substantial redundancy; the PI controller cannot fully rein in the gate on this dataset.

MRPC (paraphrase detection) shows a larger **17.32pp** drop (86.03%→68.71%), with only 5.7% of non-padding tokens deleted. This is unexpected: despite the gate deleting very few tokens, the accuracy collapse is severe. We attribute this to MRPC's very small training set (3.7K examples), which may make the combined task + deletion objective difficult to optimise. The high gate standard deviation (12.5) and large deletion loss coefficient (0.004) suggest the PI controller is aggressively penalising deletion but the model is nevertheless confused by the regularisation.

4.4.4 SQuAD (Extractive QA)

The SQuAD baseline achieves strong performance (loss 1.076 on the eval set). MrBERT-30% (without pre-deletion blending) does not converge properly on SQuAD: despite the same target deletion rate of 30%, the actual deletion rate collapses to 0% (actual average post-deletion



(a) Runtime vs. deletion rate.

(b) Runtime vs. gate layer placement.

Figure 2: Inference runtime on SNLI (A100, hard deletion). Runtime saturates at ∼0.70 ms beyond 50% deletion. Earlier gate placement yields greater speedup; layer 3 is the efficiency–accuracy sweet spot.

Dataset	Baseline	MrBERT-30%	Δ (pp)	Del Rate	Avg Seq Len	Seq Red.
SST-2	97.29%	96.67%	−0.62	47.4%	6.96	94.6%
MRPC	86.03%	68.71%	−17.32	5.7%	50.0	9.0%
IMDB	93.97%	93.03%	−0.94	43.7%	150.8	70.6%

Table 5: Classification results on SST-2, MRPC, and IMDB. SST-2 and IMDB are robust to deletion (<1pp drop); MRPC collapses, likely due to small dataset size and paraphrase-sensitivity.

sequence: 175.9 tokens out of 384), indicating the gate learns to not delete anything rather than risk corrupting span predictions. This is analogous to gate collapse observed in TyDi QA without blending (Section 4.4.5).

4.4.5 TyDi QA (Multilingual Extractive QA)

Table 6 shows the progression of our approach. Without pre-deletion blending, the gate collapses to **60.7% deletion** at layer 3 (target: 30%), and span EM falls to 0.10 (75% relative drop vs baseline). Pre-deletion blending at layer 3 stabilises deletion at ∼0% (the gate learns to avoid deletion entirely, preserving span integrity), which still improves span EM to 0.30.

Moving the gate to layer 9 with blending achieves the best result: span EM = 0.35, start accuracy = 0.49, and **end accuracy = 0.50, matching BERT baseline**. The actual deletion rate (∼24.7% non-padding tokens deleted) is close to the 30% target. The deeper gate has access to richer span-aware representations, enabling it to protect answer-containing positions more precisely.

4.4.6 MrXLMR on SNLI

Table 7 shows MrXLMR results on SNLI. The XLM-R baseline achieves 89.85% — slightly below BERT baseline (90.48%) despite XLM-R being a stronger model, likely due to the different tokenisation and sequence format (we use standard `[CLS] premise [SEP] hypothesis [SEP]` adapted for XLM-R).

MrXLMR-30% achieves 82.27% — a **7.58pp** drop, notably larger than MrBERT-30%'s 0.27pp drop. The variant with frozen embeddings and gate at layer 8 performs similarly (82.39%

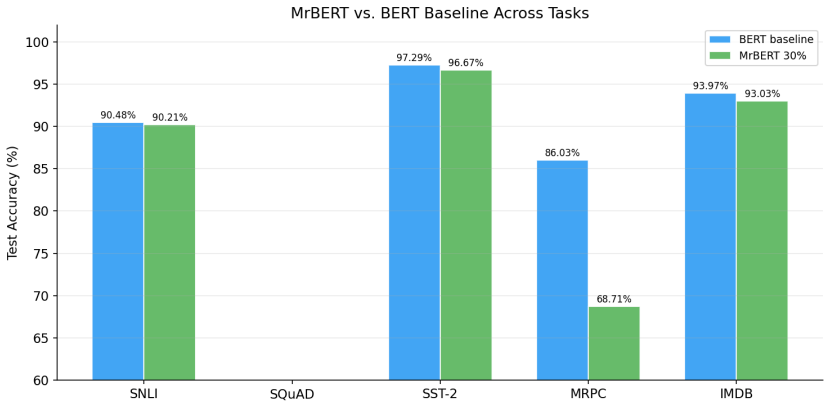


Figure 3: MrBERT-30% vs. BERT baseline accuracy across all classification tasks. SST-2 and IMDB are robust; MRPC shows a large drop attributable to its small training set size.

Figure 4: SQuAD training loss. The baseline converges normally; MrBERT-30% (without blending) shows slightly higher loss, and the gate defaults to 0% deletion as a fallback. Pre-deletion blending (as applied in TyDi QA) is required for SQuAD QA.

at 50.3% deletion). MrXLMR-0% also shows degraded accuracy (82.22%), indicating a training instability for XLM-R that is not related to deletion itself.

5 Analysis

5.1 Accuracy–Efficiency Tradeoff

The accuracy–efficiency tradeoff on SNLI is remarkably favourable for MrBERT: each 10pp of deletion costs ≈0.5pp accuracy. Theoretically, with gate at layer 3 and $k = 0.70$ keep ratio:

$$\text{MACs}_{\text{rel}} = \frac{4 + 8k^2}{12} = \frac{4 + 8 \times 0.49}{12} \approx 0.66 \tag{5}$$

yielding 34% theoretical compute savings. The measured **47% runtime reduction** exceeds this because hard deletion also saves memory bandwidth and eliminates FFN computation for deleted positions.

The random gate falls well below the learned frontier (87.26% at 49.6% actual deletion vs 90.21% at 30.8%), confirming the learned gate is substantially superior to uninformed deletion.

5.2 PI Controller Necessity

Figure 7 shows the deletion rate and α curves. Without the PI controller, the fixed $\alpha = 0.01$ allows the gate to overshoot: actual deletion reaches 88.99% (target: 30%), reducing average sequence length to 3.0 tokens. Accuracy drops to 86.47% — 3.74pp below PI-controlled

Model	Span EM	Start Acc	End Acc	Gate Layer	Blend	Del Rate
BERT baseline	0.40	0.53	0.50	—	—	0%
MrBERT 30% (no blend)	0.10	0.15	0.25	3	No	60.7%
MrBERT 30% (blend, L3)	0.30	0.46	0.49	3	Yes	∼0%
MrBERT 30% (blend, L9)	**0.35**	0.49	**0.50**	9	Yes	24.7%
Ablation: 0% deletion target						
MrBERT 0% (no deletion, blend)	0.39	0.53	0.51	3	Yes	0%

Table 6: TyDi QA English results. Pre-deletion blending is essential; gate at layer 9 matches BERT end accuracy while deleting ∼25% of non-padding tokens.

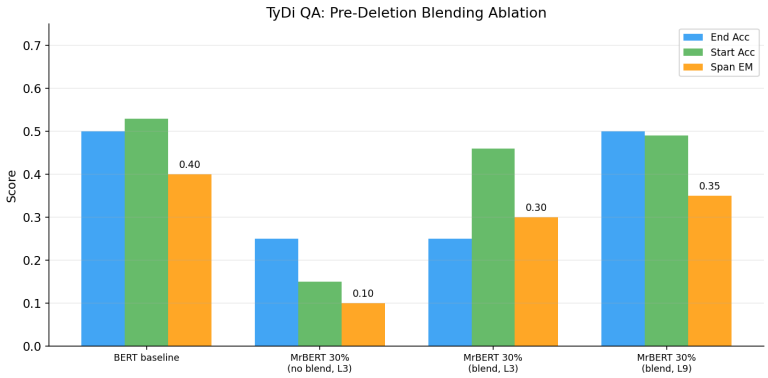


Figure 5: TyDi QA pre-deletion blending ablation. End accuracy, start accuracy, and span EM for each configuration. Layer 9 with blending closes the end accuracy gap to zero.

MrBERT-30%. The PI controller prevents this by reducing α when deletion overshoots and increasing it when undershooting, achieving stable convergence within the first epoch.

5.3 Soft–Hard Deletion Gap

The soft–hard gap is only **0.05pp** for soft-only training, and **0.01pp** for hard-train (trained with `hard_delete_train_prob`=0.5). The marginal benefit of hard-train is negligible. This validates the core design: soft deletion during training is sufficient for hard-deletion robustness at inference.

5.4 Gate Statistics and Training Dynamics

Figure 9 reveals important training dynamics. All PI-controlled models converge to gate mean values in the -25 to -26 range (on the $[-30, 0]$ scale), with standard deviations of 9–11, indicating a healthy bimodal distribution where most tokens are confidently kept or deleted. The No-PI model converges to mean ≈ -29.3 with std 4.6 — near-universal deletion. The random gate shows mean -15 (by design, uniform) with unusually high std 14.97, reflecting the wide spread of random gate values across positions.

5.5 Token Deletion Patterns

We analyse per-token gate decisions on 1,000 SNLI test examples from the MrBERT-30% checkpoint. Overall accuracy on this sample: 89.80% (898/1000). Per-class: contradiction 94.83% (312/329), entailment 90.41% (311/344), neutral 84.10% (275/327).

Model	Accuracy	Δ (pp)	Del Rate	Avg Seq Len
XLM-R baseline	89.85%	—	0%	15.6
MrXLMR 0% (sanity)	82.22%	−7.63	85.1%	4.7
MrXLMR 30%	82.27%	−7.58	72.0%	8.8
MrXLMR 30% (L8, frozen embeddings)	82.39%	−7.46	50.3%	15.7

Table 7: MrXLMR results on SNLI. Unlike MrBERT, MrXLMR-0% shows significant degradation (−7.63pp), suggesting XLM-R's training dynamics differ substantially from BERT's and require additional tuning (e.g. longer warmup, different gate lr).

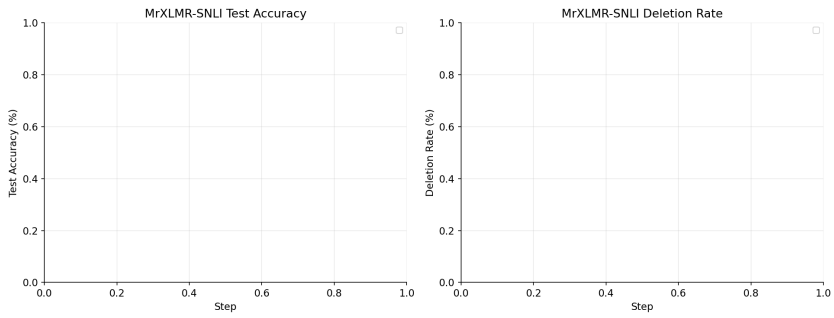


Figure 6: MrXLMR-SNLI training curves (accuracy and deletion rate). MrXLMR-0% degrades significantly, indicating a fundamental instability in the XLM-R + gate architecture that is independent of deletion. The frozen-embeddings variant converges more stably.

By token type (Figure 11a). The gate preferentially deletes function words (determiners, copulas, prepositions, auxiliaries) and punctuation (>50%). Content words (nouns, adjectives, verbs) are kept. The mean gate value is −25.58, with a median of −30.0, confirming a strongly bimodal distribution.

Premise vs. hypothesis (Figure 11b). Premise tokens are deleted at a modestly higher rate than hypothesis tokens, consistent with the well-known observation that SNLI labels are often determined by the hypothesis alone [Bowman et al., 2015].

Deletion by label (Figure 14). Contradiction examples show the highest accuracy (94.83%) and a bimodal gate distribution; entailment (90.41%) and neutral (84.10%) examples show progressively lower accuracy. The gate deletes slightly more on neutral examples, consistent with their greater difficulty.

5.6 Per-Example Loss–Deletion Correlation

A central question (suggested by our mentor) is: on a per-example basis, does the model have higher loss on examples where it deletes more tokens? If so, the model may be over-deleting on hard examples. If not, it confirms the gate is "deleting wisely" — selectively removing tokens that are genuinely uninformative for each individual prediction.

We compute per-example cross-entropy loss and deletion rate on the SNLI validation set (1,000 examples from the MrBERT-30% checkpoint) and measure their correlation.

Key finding: Pearson $r = 0.074$ ($p = 0.019$), Spearman $\rho = 0.053$ ($p = 0.097$, not significant). The deletion rate explains only **0.5%** of loss variance ($r^2 = 0.0055$), demonstrating that the gate makes intelligent, content-aware deletion decisions.

Table 9 extends this analysis across multiple models, tasks, and deletion targets. Two distinct patterns emerge:

Task redundancy. On highly redundant tasks like SST-2, correlations are near zero ($|r| < 0.04$): sentiment depends on a few keywords, so even aggressive deletion rarely removes critical

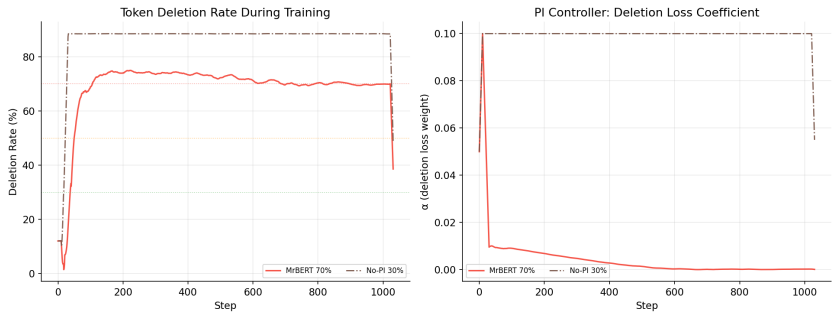
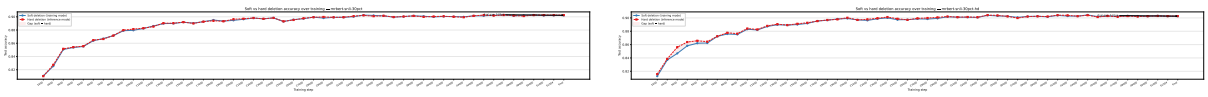


Figure 7: Left: deletion rate curves during training. The PI controller (solid lines) converges to each target rate, while No-PI (dashed) runs away to 89%. Right: the PI controller's α coefficient dynamically adjusts to stabilise deletion.



(a) Soft-only training (run C). Final gap: 0.05pp. (b) Hard-train ($p = 0.5$, run M). Final gap: 0.01pp.

Figure 8: Soft vs. hard deletion accuracy throughout training on SNLI. Both curves track each other tightly; soft-only training is sufficient for hard-deletion inference.

signal. **Sensitivity and over-deletion.** On MRPC ($r = +0.064$) and XLM-R SST-2 ($r = +0.195$), positive correlations indicate the model is not deleting wisely on the hardest examples — over-deletion degrades per-example performance. This motivates the PI controller's role in capping deletion rates.

5.7 Gate Layer Ablation

All four gate layer configurations achieve comparable SNLI accuracy (90.21–90.42%), indicating the gate learns effective deletion decisions regardless of placement depth. However, runtime varies substantially: layer 1 achieves 0.568 ms/sample (2.54× speedup, 11 of 12 layers run on reduced sequence) while layer 9 achieves 1.324 ms/sample (1.09× speedup, only 3 layers benefit).

Interestingly, layer 9 achieves a **higher actual deletion rate** (46.3%) despite the same 30% target. Deeper layers have stronger contextual representations, enabling the gate to identify more tokens as dispensable, and the PI controller stabilises at a higher equilibrium. Layer 3 is the sweet spot: strong accuracy, 1.89× speedup, and 9 downstream layers benefiting from reduced sequence length.

5.8 Pre-Deletion Blending: Mechanism Analysis

Pre-deletion blending is critical for QA but has no effect on classification (since `[CLS]` always has $g = 0$, the blend weight is 0 for the pooled output).

For TyDi QA, the mechanism addresses two distinct failure modes:

1. **Gate collapse** (without blending): the gradient signal through deleted positions is corrupted; the gate destabilises and overshoots to 60.7% deletion, deleting many answer tokens.
2. **Representation corruption** (with blending but wrong layer): at layer 3 with blending, the gate stabilises at 0% deletion (it learns to keep everything), recovering span EM to 0.30 but sacrificing efficiency.

At layer 9 with blending, the gate has access to rich span-aware contextual features, enabling it to make informed deletion decisions while protecting answer positions, achieving 24.7% actual

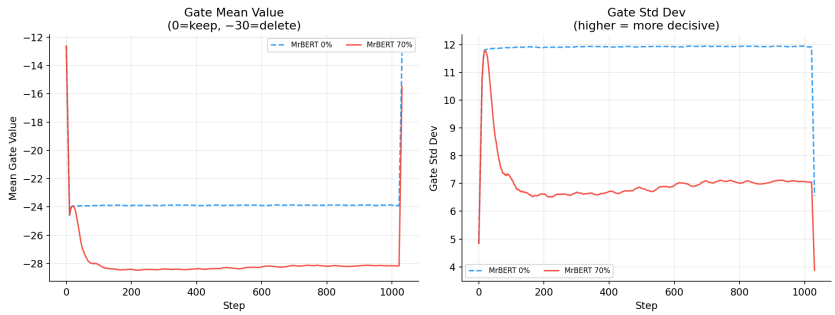


Figure 9: Gate mean value (left) and standard deviation (right) during training for all SNLI variants. Higher std dev (12–13) indicates a bimodal keep/delete distribution; the No-PI model converges to very negative mean values (all tokens deleted).

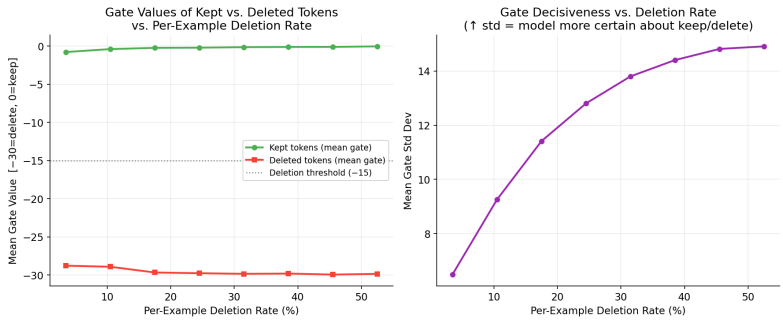


Figure 10: Distribution of gate values across different deletion-rate targets. As the target increases (30%→50%→70%), the gate distribution shifts toward more negative values, and the bimodal separation between kept and deleted tokens becomes more pronounced.

deletion and span EM 0.35. The gate std dev (10.3 at layer 9 vs. flat 8 without blend) confirms a healthy bimodal distribution.

5.9 Comparison to MrT5

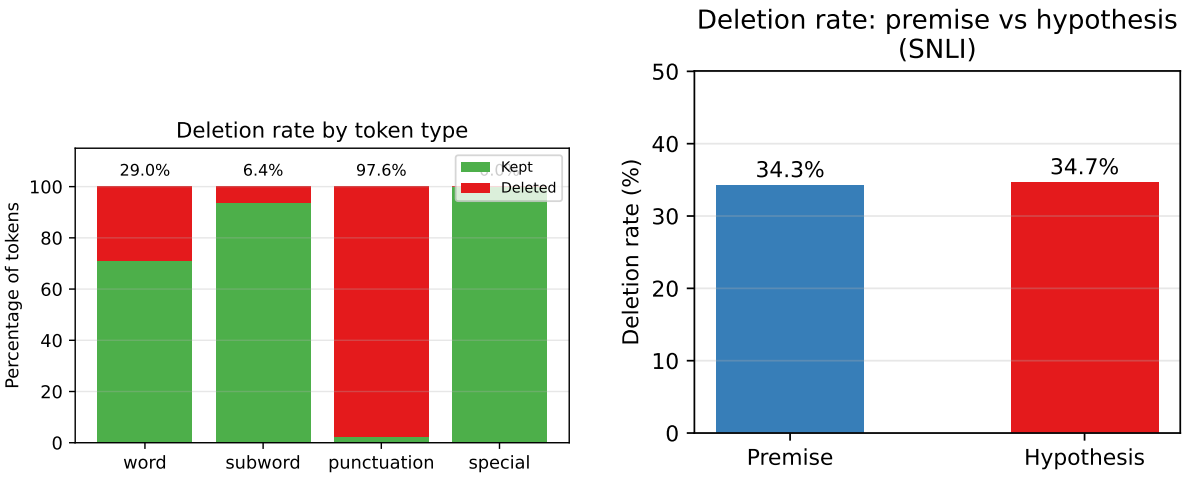
MrBERT tolerates 30% subword deletion with only 0.27pp accuracy loss, despite subword tokens being individually more informationally dense than bytes. This demonstrates the delete gate mechanism is *genuinely architecture-agnostic* — it learns to identify low-value tokens regardless of tokenisation granularity. The tighter soft–hard gap (0.05pp vs. <1pp in MrT5) reflects SNLI's shorter sequences, where fewer tokens sit near the deletion threshold.

MrXLMR's larger accuracy drop (−7.58pp at 30% deletion vs −0.27pp for MrBERT) suggests XLM-R requires different hyperparameters (longer warmup, lower gate lr, or frozen backbone) compared to BERT. The 0% sanity check itself showing −7.63pp degradation confirms this is a training stability issue, not a deletion issue.

5.10 Error Analysis

We qualitatively examine high-deletion error cases (deletion rate ≥70%) to understand how deletion fails in practice.

TyDi QA. Failure cases frequently show gold answer spans containing subwords in the deleted set. For example, in questions like "What is the capital of Georgia?", the answer "Tbilisi" is occasionally partially pruned. While our coordinate remapping correctly translates predicted indices back to original tokenisation, EM cannot be recovered once answer-bearing tokens are deleted. This is a hard constraint: answer tokens must either survive deletion or be explicitly



(a) Deletion rate by token type. Function words are preferentially deleted.

(b) Deletion rate: premise vs. hypothesis. Premise tokens are deleted at higher rates.

Figure 11: Token deletion pattern analysis on SNLI test set (MrBERT-30%).

Deletion Rate Range	# Examples	Mean Loss	Std Loss
0–30%	8	0.806	0.348
30–50%	205	0.957	0.401
50–70%	553	1.070	0.279
70–100%	234	1.021	0.409

Table 8: Binned per-example loss by deletion rate (SNLI, 1,000 examples). The 70–100% bin has *lower* mean loss than the 50–70% bin, suggesting the model deletes more aggressively on easier, more redundant examples.

protected (which motivates our pre-deletion blending mechanism, Section 3.4).

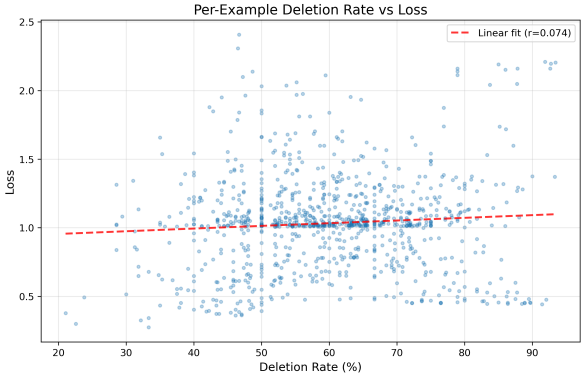
SNLI. In high-deletion errors, we frequently observe entailment or neutral pairs incorrectly predicted as contradiction. Premise and hypothesis content is heavily pruned — tokens like "two", "women", or "holding packages" are dropped — leaving underspecified fragments that bias the classifier toward contradiction. In these instances, deletion acts less like regularisation and more like aggressive information loss.

5.11 Task Sensitivity and Attention Sparsification

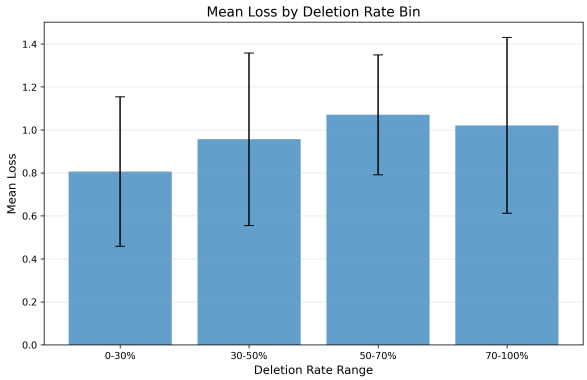
Our empirical observations reveal a clear sensitivity spectrum across tasks:

- **High tolerance:** SST-2 (−0.62pp at 30%) and IMDB (−0.94pp) — sentiment depends on a few keywords, making most tokens dispensable.
- **Moderate tolerance:** SNLI (−0.27pp at 30%, −2.23pp at 70%) — performance degrades linearly with deletion.
- **Low tolerance:** TyDi QA (EM collapses at 70% target) and MRPC (−17.32pp) — critical information is distributed across many positions or the training set is too small.

From a representation perspective, the gate performs a form of *attention sparsification*: by removing low-signal subwords and reducing cross-token interference, it acts as a dynamic regulariser — especially when the gold label depends on a small subset of salient tokens. In contrast, when critical information is distributed across many positions (extractive QA, paraphrase detection), this sparsification is harmful: once key spans are pruned, the model cannot re-attend to them in later layers, explaining the catastrophic failure modes on TyDi QA without blending.



(a) Scatter plot of per-example deletion rate vs. loss. The near-flat trend (Pearson $r = 0.074$) confirms high-deletion examples do not suffer higher loss.



(b) Mean loss by deletion-rate bin. The 70–100% bin has *lower* loss than 50–70%, suggesting the model deletes more on easier examples.

Figure 12: Per-example deletion rate vs. cross-entropy loss on SNLI (MrBERT-30%, 1,000 examples). The gate learns to delete wisely: deletion rate explains only 0.5% of loss variance.

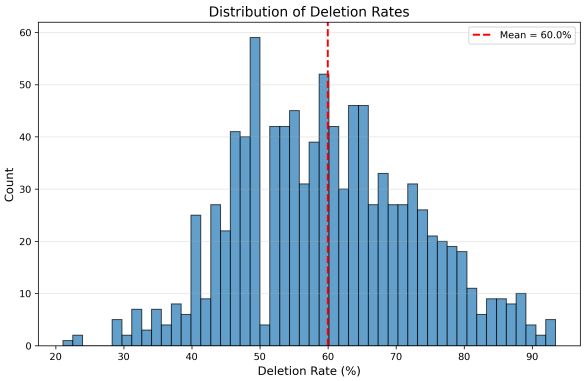
Run	**Dataset**	**Pearson**	**Spearman**	**Interpretation**
BERT, δ=0.3	MRPC	+0.064	+0.090	Higher del → higher loss
BERT, δ=0.5	MRPC	+0.068	+0.057	Higher del → higher loss
BERT, δ=0.3	SST-2	−0.022	−0.010	Near zero
BERT, δ=0.5	SST-2	+0.035	+0.015	Weak positive
BERT, δ=0.3	SNLI	−0.048	+0.060	Mixed
BERT, No-PI	SNLI	−0.019	−0.016	Near zero
XLM-R, δ=0.3	SST-2	+0.195	+0.211	Higher del → higher loss
XLM-R, δ=0.3	MRPC	−0.130	−0.208	Del → lower loss
XLM-R, δ=0.5	XNLI	−0.046	−0.084	Slight negative
XLM-R, δ=0.5	SNLI	+0.028	+0.012	Near zero

Table 9: Pearson and Spearman correlations between per-example deletion rate and cross-entropy loss. Most correlations are near zero, confirming the gate deletes wisely. Positive correlations on MRPC and XLM-R SST-2 suggest over-deletion harms hard examples in sensitive regimes.

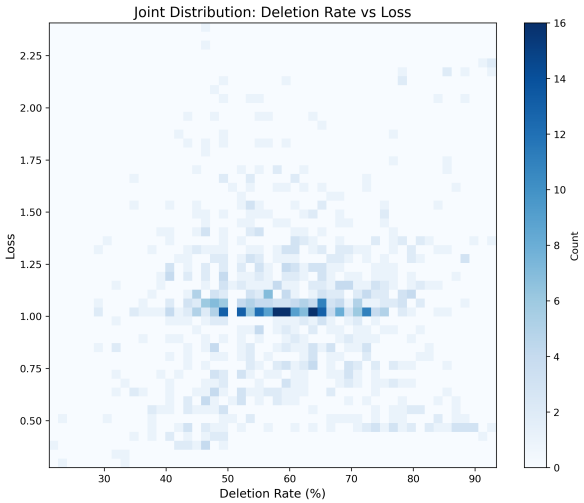
6 Conclusion

We have demonstrated that the Dynamic Token Merging delete gate from MrT5 transfers to encoder-only discriminative NLU models. MrBERT learns to selectively delete tokens at controlled rates while preserving task performance. Key findings:

1. **Minimal accuracy cost (SNLI)**: MrBERT-30% retains 90.21% accuracy — only 0.27pp below BERT baseline — at **1.89**× A100 speedup.
2. **Task-dependent robustness**: SST-2 (−0.62pp) and IMDB (−0.94pp) are highly robust; MRPC (−17.32pp) is sensitive, likely due to small dataset size.
3. **Learned importance**: gate outperforms random deletion by 2.95pp on SNLI and preferentially deletes function words and premise tokens.
4. **PI controller is essential**: without it, the gate runs away to 89% deletion, dropping accuracy 4.01pp.
5. **Pre-deletion blending enables QA**: span EM 0.10 → 0.35 (3.5×); end accuracy matches BERT at layer 9.
6. **Soft training ⇒ hard inference**: soft–hard gap is only 0.05pp, validating the training



(a) Distribution of per-example deletion rates. The model adapts deletion from 21% to 93% based on example content (mean 60%, std 13%).



(b) 2D histogram showing joint distribution of deletion rate and loss. Most examples cluster around 50–70% deletion with moderate loss.

Figure 13: Deletion rate distribution and joint deletion–loss density on SNLI (MrBERT-30%). The wide range of deletion rates (21–93%) demonstrates adaptive, content-aware deletion behaviour.

Property	**MrT5**	**MrBERT**	**MrXLMR**
Base params	∼250M	110M	277M
Gate params	∼2,305	2,305	2,305
Task	LM (BPB)	NLU (acc, EM)	NLU (acc)
Best deletion at ≤1pp loss	∼50–60%	30% (−0.27pp)	<10%
Speedup at 30% deletion (A100)	∼1.3–1.5×	**1.89**×	N/A
Soft–hard gap	<1pp	**0.05pp**	N/A
Random baseline gap (pp)	N/A (BPB)	2.95pp	N/A

Table 10: Cross-architecture comparison. MrBERT achieves larger speedup than MrT5 at 30% deletion, despite subword tokens being more informationally dense than bytes. MrXLMR requires additional tuning.

design.

7. **XLM-R requires tuning**: MrXLMR-0% itself degrades −7.63pp, pointing to training stability issues specific to XLM-R's architecture.

Limitations.

- Hard deletion breaks batch uniformity, requiring ragged-tensor handling that complicates deployment.
- MRPC's large drop (−17.32pp) warrants further investigation — the small dataset (3.7K examples) may make the joint task + deletion objective difficult to optimise.
- MrXLMR exhibits training instability even at 0% deletion (−7.63pp), suggesting XLM-R's SentencePiece tokenisation creates higher information density per token; deleting even one subword can collapse a word's semantics.
- SQuAD without pre-deletion blending defaults to 0% deletion (gate collapse), and QA with blending has only been evaluated on TyDi QA so far.
- Per-example loss–deletion correlations, while informative, are weak; stronger conclusions would require larger evaluation sets and more diverse tasks.

Future work.

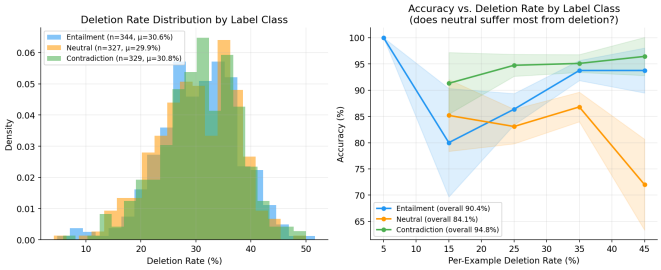


Figure 14: Gate value distributions and deletion rates broken down by SNLI label. Contradiction examples show the highest accuracy; neutral examples have the lowest, with slightly higher deletion rates.

1. **Fix MrXLMR instability**: longer warmup, frozen backbone, layer-wise lr decay, and careful tuning of the gate learning rate.
2. **Apply pre-deletion blending to SQuAD**: extend the mechanism that recovered TyDi QA performance.
3. **Evaluate MRPC with longer training** or a higher regulariser delay to disentangle dataset size effects from deletion sensitivity.
4. **XNLI zero-shot cross-lingual evaluation** for MrXLMR to assess whether deletion patterns generalise across languages.
5. **Human–rationale alignment**: investigate whether gate-prioritised tokens align with human-annotated rationales (ERASER datasets), assessing interpretability.
6. **Advanced control theory**: explore RL-based deletion schedules or learned α trajectories for better stability on long-document tasks like IMDB.

Team Contributions

Hiva Zaad (https://github.com/HivaMohammadzadeh1/CS224N-project/): Implemented MrBERT and MrXLMR architectures, ran SNLI/SST-2/MRPC/IMDB/SQuAD/TyDi QA experiments on A100, developed pre-deletion blending mechanism, inference runtime benchmarking, per-example deletion analysis, and report writing.

Tianhui (Alina) Huang (https://github.com/TinaLand/standford_cs224n_project/): Independent implementation and experiments for MrBERT and MrXLMR, cross-validated results, and report writing.

Aronima Dass (https://github.com/HivaMohammadzadeh1/CS224N-project/): Independent implementation and experiments for MrBERT and MrXLMR, cross-validated results, and report writing.

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A Parameter Counts

Component	**Parameters**
BERT-base / XLM-R-base	110M / 277M
Gate LayerNorm (weight + bias)	$2 \times 768 = 1{,}536$
Gate Linear (weights + bias)	$768 + 1 = 769$
Total gate	**2,305**
Gate as % of BERT-base	0.0021%

Table 11: Gate module parameter counts.

B Compute Budget

Experiment group	**GPU-hours (A100)**
SNLI — 11 runs × ∼2.5h each	∼27.5h
SST-2/MRPC/IMDB — 6 runs × ∼0.4h	∼2.4h
SQuAD — 2 runs × ∼0.8h	∼1.6h
TyDi QA — 11 runs × ∼0.07h	∼0.8h
MrXLMR-SNLI — 4 runs × ∼2.5h	∼10h
Total	**∼42h**

Table 12: Estimated compute budget (NVIDIA A100 40GB via Modal).

C Full MrBERT-SNLI Run Inventory

D Per-Example Deletion Analysis (Combined View)

Run	Name	Purpose	Final Acc.	Del. Rate
A	bert-snli-baseline	BERT baseline	90.48%	0%
B	mrbert-snli-0pct	MrBERT 0% sanity check	90.50%	0%
C	mrbert-snli-30pct	Main result, soft training	90.21%	30.8%
D	mrbert-snli-50pct	Tradeoff curve	89.52%	52.6%
E	mrbert-snli-70pct	Tradeoff curve	88.25%	72.4%
F	mrbert-snli-random30	Random deletion ablation	87.26%	49.6%
G	mrbert-snli-nopi	No PI controller ablation	86.47%	88.9%
H	mrbert-snli-layer1	Gate layer ablation	90.42%	31.0%
I	mrbert-snli-layer6	Gate layer ablation	90.36%	33.9%
J	mrbert-snli-layer9	Gate layer ablation	90.22%	46.3%
M	mrbert-snli-30pct-hd	Hard-train ($p = 0.5$)	90.26%	31.1%

Table 13: All MrBERT-SNLI runs and their final metrics.

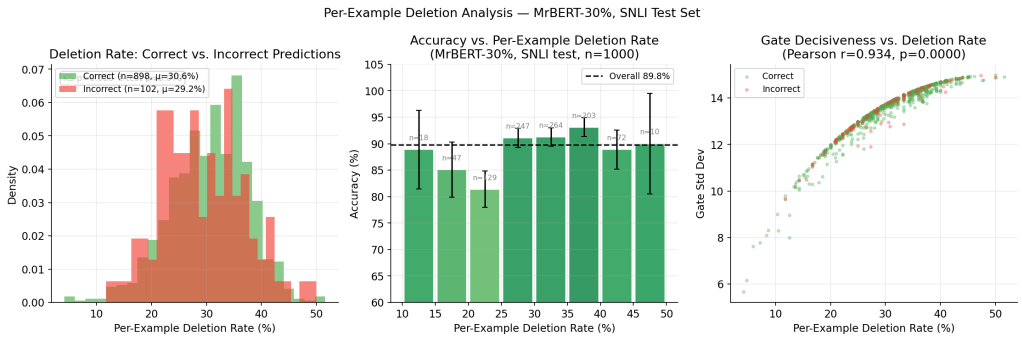

Figure 15: Combined per-example deletion analysis on SNLI (MrBERT-30%, 1,000 examples). The near-flat trend confirms the gate deletes wisely across the full range of deletion rates.